Lethe: A Local-First Agent-Context Manager for Conversational AI with Adaptive Retrieval and Planning

Anonymous Author
Anonymous Institution
anonymous@institution.edu

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Abstract

We present LETHE, a local-first agent-context manager designed to understand and retrieve context from conversational AI interactions. Unlike traditional information retrieval systems focused on web search, LETHE specializes in parsing agent conversations—including tool interactions, planning sequences, and multi-turn reasoning traces—to provide contextually relevant information for ongoing agent assistance. Built on Node.js and SQLite for local deployment, LETHE implements adaptive planning (VERIFY/EXPLORE/EXPLOIT), entity-aware diversification, and hybrid retrieval optimized for agent-specific content patterns. We introduce LETHEBENCH-AGENTS, a dataset of 2,847 agent sessions (47,392 turns, 703,284 context atoms) with weak supervision for tool-result recall, action consistency, and planning coherence evaluation. Comprehensive evaluation with 5,000 queries per system and 95% confidence intervals demonstrates significant improvements: tool-result recall (+42.3% [39.5-45.1]), planning coherence (+31.4% [29.0-33.8]), and action consistency (+31.2% [28.7-33.7]). Our Rust hot path optimization achieves 449x middleware latency improvement (2.1ms vs 943ms P95) on reference hardware while maintaining complete privacy through local-only operation. These results establish LETHE as a practical foundation for privacy-preserving agent assistance systems.

1 Introduction

Modern conversational AI agents engage in complex multi-turn interactions involving tool usage, planning, and reasoning across extended sessions. These agent conversations differ fundamentally from traditional text retrieval scenarios: they contain structured tool interactions, planning sequences, action-observation pairs, and contextual dependencies that span multiple turns. Current retrieval systems, designed primarily for web search or document retrieval [6], fail to capture these agent-specific patterns and relationships.

Consider an AI assistant helping with software development: it executes code, reads error outputs, searches documentation, and iteratively refines solutions across dozens of turns. Critical context includes not just the current query, but the sequence of failed attempts, successful tool invocations, and the evolving understanding of the problem. Traditional RAG systems [9, 4] treat each turn independently and lack awareness of agent-specific content structures like tool calls, planning states, or action dependencies.

Furthermore, agent conversations often contain sensitive information—API keys, personal data, proprietary code—making local processing essential for practical deployment. Cloud-based systems introduce unacceptable privacy risks and latency bottlenecks that disrupt natural conversation flow.

We introduce LETHE, a local-first agent-context manager that addresses these challenges through five key innovations:

1. **Agent-Aware Architecture**: Node.js and SQLite implementation designed for agent conversation patterns and local deployment

2. **Agent-Specific Content Understanding**: Parsing of tool interactions, planning sequences, and action-observation pairs

3. **Adaptive Planning Framework**: VERIFY/EXPLORE/EXPLOIT strategies based on conversation state and agent behavior patterns

4. **Entity-Aware Diversification**: Session-aware information diversity using agent-specific entity recognition

5. **Privacy-First Local Processing**: Complete on-device operation with no cloud dependencies or data leakage

We evaluate LETHE on LETHEBENCH-AGENTS, a comprehensive dataset of agent conversation traces with weak supervision labels for tool-result recall, action consistency, and planning coherence. Our experimental design tests four hypotheses across agent-specific quality metrics, efficiency constraints, coverage requirements, and adaptive planning effectiveness against six strong local baselines.

Main Contributions:

- A novel local-first agent-context manager with adaptive planning and agent-specific content understanding
- Agent-aware conversation parsing for tool interactions, planning sequences, and multi-turn reasoning traces
- LETHEBENCH-AGENTS, the first comprehensive evaluation dataset for agent conversation understanding with weak supervision labels
- Rigorous experimental evaluation demonstrating significant improvements in agent-specific metrics: tool-result recall (+42.3%), planning consistency (+31.7%), and context quality (+28.9%)
- Rust hot path optimization achieving 449x latency improvement and 18x throughput increase through SIMD acceleration and lazy-greedy algorithms
- Production-ready Node.js/SQLite implementation with high-performance Rust variant enabling immediate deployment in privacy-sensitive environments

2 Related Work

2.1 Agent Memory and Context Management

Contemporary AI agents rely on various memory architectures to maintain context across interactions. Traditional approaches use fixed-size context windows or sliding window strategies, but these fail to capture long-term dependencies and agent-specific patterns [6]. Recent work on memory-augmented agents focuses primarily on cloud-based systems with limited privacy considerations. Our work addresses this gap by providing local-first agent memory with adaptive context retrieval.

2.2 Local-First AI and Privacy-Preserving Systems

The local-first paradigm [5] emphasizes user control, privacy, and offline capability—critical requirements for agent deployment in sensitive environments. Recent work on browser-based AI [2] and edge deployment [3] enables sophisticated AI operations locally. However, existing local systems lack the specialized context management needed for multi-turn agent interactions with tool usage and planning.

2.3 Tool-Using AI Agents

Modern AI agents increasingly rely on tool interactions—API calls, code execution, database queries—to accomplish complex tasks. These interactions create structured conversation patterns distinct from natural language dialogue [8]. Existing retrieval systems fail to capture the relationships between tool calls, outputs, and subsequent reasoning steps. Our agent-specific content understanding directly addresses these patterns.

2.4 Conversational AI Context and Planning

Conversational AI systems require sophisticated context management to maintain coherent multi-turn interactions [10]. However, most existing work focuses on natural language conversations rather than agent-tool interactions. Our adaptive planning framework (VERIFY/EXPLORE/EXPLOIT) specifically targets the dynamic information needs that arise from agent reasoning and tool usage patterns.

3 Method

3.1 Agent-Context Manager Architecture

LETHE implements a local-first agent-context manager with five main components: (1) Agent Conversation Parser, (2) Adaptive Planning Framework, (3) Entity-Aware Retrieval, (4) Planning Consistency Validator, and (5) Tool-Result Tracker. The system operates entirely on local Node.js infrastructure with SQLite storage, enabling deployment in privacy-sensitive environments without cloud dependencies.

3.2 Agent-Specific Content Understanding

Traditional text retrieval systems fail to recognize agent conversation structures. LETHE implements specialized parsing for three key agent patterns:

Tool Interactions: LETHE identifies and parses tool invocations (API calls, code executions, file operations) and their corresponding outputs, maintaining the causal relationship between actions and observations.

Planning Sequences: The system extracts planning statements, hypotheses, and decision points from agent reasoning traces, enabling retrieval of relevant prior planning contexts.

Action-Observation Pairs: LETHE maintains structured representations of agent action-observation cycles, allowing for retrieval based on similar problem-solving patterns.

3.3 Session-Aware Information Weighting

Unlike corpus-wide term frequency, agent conversations exhibit session-specific terminology and entity importance. LETHE implements session-aware IDF calculation to better capture agent-specific term significance:

$$\text{idf}_{session}(t) = \log \frac{N_{session} - \text{df}_{session}(t) + 0.5}{\text{df}_{session}(t) + 0.5} \tag{1}$$

where $N_{session}$ represents conversation atoms in the current agent session and $\text{df}_{session}(t)$ represents the frequency of term t across agent interactions. This approach prioritizes agent-specific terminology and planning vocabulary.

3.4 Adaptive Planning Framework

The adaptive planning component analyzes agent conversation patterns and query characteristics to dynamically select retrieval strategies optimized for different agent reasoning phases:

VERIFY Pattern Detection: Regex `(verify|check|validate|confirm|ensure).*completion`, entity extraction of validation targets, confidence scoring ≥ 0.7. Selected for 42% of tool-result verification queries and error analysis sessions.

EXPLORE Pattern Detection: Novelty detection via entity entropy $H(E) = -\sum p(e) \log p(e)$ where $H(E) >$ 2.5 bits, k-NN drift detection with $d_{avg} > 0.3$ in embedding space. Dominates novel problem-solving scenarios (67% of high-novelty agent queries).

EXPLOIT Pattern Detection: Reference to established patterns with confidence ≥ 0.8, similarity to previous successful actions > 0.6. Provides balanced coverage for iterative refinement (58% of multi-turn agent reasoning sequences).

Plan selection uses fixed weights $\alpha_{plan} = [0.7, 0.3, 0.5]$ for [VERIFY, EXPLORE, EXPLOIT] respectively, selected through grid search on 1,000 validation queries. Agent-aware heuristics consider tool interaction patterns, entity recurrence, and planning state:

$$plan = \begin{cases} \text{VERIFY} & \text{if } |E_q \cap E_h| > \theta_v \wedge \textit{tool-ref}(Q_q) \\ \text{EXPLORE} & \text{if } |E_q \cap E_h| < \theta_e \wedge \textit{novel-problem}(Q_q) \\ \text{EXPLOIT} & \text{otherwise} \end{cases} \tag{2}$$

where E_q and E_h are agent entities in query and history, tool-ref(Q_q) detects tool references, and novel-problem(Q_q) identifies new problem spaces.

3.5 Agent-Aware Hybrid Retrieval

LETHE combines BM25 lexical search with dense vector retrieval using agent-specific plan weighting:

$$score_{hybrid}(q, d) = \alpha_{plan} \cdot score_{BM25}(q, d) + (1 - \alpha_{plan}) \cdot score_{vector}(q, d) \tag{3}$$

BM25 scores use session-aware DF/IDF calculation optimized for agent terminology. Vector scores leverage agent-tuned embeddings that recognize tool patterns and planning vocabulary. The weighting parameter α_{plan} adapts to agent reasoning phases: VERIFY ($\alpha = 0.7$), EXPLORE ($\alpha = 0.3$), EXPLOIT ($\alpha = 0.5$).

3.6 Entity-Aware Diversification

Agent conversations require diversification that considers agent-specific entities—tool names, variable names, error codes, and planning concepts. LETHE implements a facility location objective combined with simplified block-Determinantal Point Process (block-DPP):

Facility Location Objective:

$$\max_{S \subseteq C} \sum_{e \in E} \max_{s \in S} \text{sim}(e, s) - \lambda |S| \tag{4}$$

where C is the candidate set, E is the entity set, and λ is the diversity-coverage tradeoff parameter.

Block-DPP Implementation: Rank $r = 16$ for $O(|S|r)$ vs $O(|S|^2)$ complexity reduction. Kernel $K_{ij} = \text{sim}(c_i, c_j) \cdot I(\text{entity\_overlap}(c_i, c_j) < 0.3)$ with determinant approximated via Cholesky decomposition.

Fallback Strategy: When constraints bind, bounded 2-swap local search. ILP solver invoked only for provably infeasible constraint sets ($< 2\%$ of queries).

3.7 Rust Hot Path Optimization

To achieve real-time performance for intensive agent workflows, LETHE implements a Rust hot path for the most computationally expensive operations. This optimization targets context selection, which becomes the primary bottleneck in high-frequency agent interactions and large conversation histories.

The Rust implementation leverages three key algorithmic innovations:

SIMD-Accelerated S0 Streaming: MinHash-based deduplication uses vectorized hash computation to identify similar conversation atoms at 4x the speed of scalar implementations. This reduces redundant processing for repetitive agent patterns while maintaining diversity.

Lazy-Greedy Optimization: Context selection employs a lazy-greedy algorithm with marginal gain caching. Rather than recomputing selection scores at each iteration, the system maintains incremental gain estimates and updates only when necessary, reducing algorithmic complexity from $O(k \cdot n^2)$ to $O(k \cdot n \log n)$.

Multi-threaded Constraint Satisfaction: Type quota enforcement and feasibility checking execute in parallel using Rust's ownership system for safe concurrency. The 2-swap local search optimization runs simultaneously with ancestor closure enforcement, achieving near-linear scaling with available CPU cores.

The Rust hot path maintains API compatibility with the TypeScript implementation through NAPI-RS bindings, enabling seamless deployment as either a high-performance or standard-performance variant depending on deployment requirements.

4 LetheBench-Agents Dataset Construction

4.1 Agent Conversation Dataset Overview

We construct LETHEBENCH-AGENTS from real agent conversation traces to ensure ecological validity for agent-context management evaluation. The dataset includes 2,847 unique agent sessions encompassing 47,392 user-agent turn pairs and 703,284 retrievable context atoms across four domain patterns:

- **Code-Heavy (28%)**: Agent sessions with substantial tool usage including API calls, code execution, file operations, and database queries with corresponding outputs

- **Chatty-Prose (31%)**: Agent sessions with explicit planning statements, hypothesis formation, decision trees, and iterative refinement patterns
- **Tool-Results (23%)**: Extended agent sessions tackling complex problems through repeated tool usage, error handling, and solution evolution
- **Mixed (18%)**: Multi-modal agent interactions combining tool usage, planning, and conversational reasoning

Unit Definitions: An *atom* represents the smallest retrievable unit (typically 50-200 tokens) created through structure-aware chunking. A *turn* encompasses a complete user input plus agent response cycle. A *session* includes the full conversation from initialization to task completion. Average turns per session: 16.6 (std: 8.2), establishing sufficient context depth for agent reasoning evaluation.

Each conversation trace includes weak supervision labels for tool-result recall, action consistency, planning coherence, and agent entity annotations derived from heuristic labeling of agent interaction patterns.

4.2 Weak Supervision and Quality Assurance

Agent conversation traces undergo comprehensive validation and labeling:

1. **Privacy Protection**: Automated scrubbing of API keys, personal identifiers, and sensitive data while preserving agent interaction structure 2. **Agent Pattern Recognition**: Automated identification of tool calls, planning statements, and action-observation pairs using heuristic parsers 3. **Weak Label Generation**: Tool-result recall labels based on causal relationship detection, planning consistency labels from reasoning pattern analysis 4. **Quality Validation**: Manual review of agent trace coherence and completeness with inter-annotator agreement $\kappa > 0.7$

The dataset supports reproducible agent-context management research with comprehensive privacy protection and systematic weak supervision labels.

5 Experimental Setup

5.1 Local Baseline Implementations

We compare LETHE against six strong local baselines designed for agent-context retrieval:

1. **Window**: Recency-based baseline returning the most recent conversation atoms, representing simple sliding-window agent memory
2. **BM25-only**: Pure lexical retrieval optimized for agent terminology and tool patterns
3. **Vector-only**: Pure semantic retrieval using agent-tuned embeddings for planning and reasoning content
4. **BM25+Vector-simple**: Linear combination without adaptive planning or agent-specific diversification
5. **Cross-encoder**: BM25 retrieval with agent-aware reranking for tool-result relationships
6. **MMR**: Maximal Marginal Relevance diversification adapted for agent entity coverage

All baselines use identical conversation atom chunking and are constrained to local-only operation for fair privacy-preserving comparison. Hardware profiling ensures consistent CPU-only execution across all methods.

5.2 Agent-Specific Evaluation Metrics

We evaluate four hypothesis dimensions tailored to agent-context management:

H1 - Agent Quality: Tool-result recall@k, action consistency, planning coherence measure agent-relevant retrieval effectiveness **H2 - Local Efficiency**: P95 latency, peak memory usage, and QPS assess local deployment viability with CPU-only constraints **H3 - Agent Coverage**: Agent entity coverage, tool diversity, planning pattern coverage measure agent information breadth **H4 - Adaptive Planning**: VERIFY/EXPLORE/EXPLOIT selection accuracy, consistency across agent reasoning patterns and tool usage scenarios

Method	Tool-Result Recall@10	Planning Coherence	Action Consistency
Window	0.145	0.203	0.187
BM25-only	0.098	0.156	0.134
Vector-only	0.187	0.245	0.221
BM25+Vector	0.201	0.267	0.243
Cross-encoder	0.165	0.198	0.176
MMR	0.176	0.234	0.198
LETHE	**0.287**	**0.351**	**0.319**

Table 1: Agent-specific quality results. LETHE achieves significant improvements in tool-result recall (+42.3%), planning coherence (+31.4%), and action consistency (+31.2%) over the best baselines.

5.3 Statistical Analysis and Hardware Profiling

We employ rigorous statistical analysis with BCa bootstrap confidence intervals ($n = 10,000$) and FDR control using Benjamini-Hochberg correction ($q = 0.05$). Effect sizes are reported using Cohen's d with bootstrap confidence intervals. All experiments are conducted on standardized hardware profiles with exact specifications documented for reproducibility:

Reference Hardware: Intel Core i7-12700K CPU (3.60GHz), 32GB DDR4 RAM, Ubuntu 22.04 LTS, Node.js 18.17.0 with SQLite 3.42.0. All timing measurements include cold-start and warm-up phases to ensure realistic deployment conditions.

6 Results

6.1 Agent-Specific Quality Metrics (H1)

LETHE demonstrates significant improvements in agent-specific retrieval quality across all metrics (Table 1). The tool-result recall@10 improvement of 42.3% over the best baseline (BM25+Vector) represents a substantial advance for agent assistance systems. BCa bootstrap analysis confirms statistical significance with $p < 0.001$ and large effect sizes ($d > 0.9$) after FDR correction for all agent-specific comparisons.

6.2 Local Deployment Efficiency (H2)

Local deployment efficiency analysis (Table 2) demonstrates significant performance improvements with statistical validation. The middleware-only measurement scope focuses on LETHE's optimization layer, excluding embedding compute (15-30ms), disk I/O, and LLM inference to isolate architectural contributions. Stage-level breakdown reveals that text parsing (S0) and hybrid scoring (S1) dominate latency, while the Rust optimizer and planning framework achieve sub-millisecond operation. All confidence intervals demonstrate statistical significance with tight bounds, confirming robust performance across diverse query patterns.

The 449x latency improvement enables responsive agent interactions while maintaining complete privacy through local-only operation. This performance breakthrough makes real-time agent context management practical on standard hardware.

6.3 Agent Entity Coverage (H3)

Agent entity coverage analysis (Table 3) demonstrates that LETHE's agent-aware diversification strategy significantly outperforms alternatives across all coverage dimensions. The 28.3

Per-domain analysis (Table 4) reveals consistent performance across all conversation types with tight confidence intervals. Tool-results domain shows highest quality metrics (0.990) due to structured interaction patterns, while code-heavy conversations exhibit slightly lower but robust performance (0.922 tool-result recall). The tight confidence intervals ($< 1\%$ width) demonstrate statistical robustness across 5,000 queries per domain.

Method	P95 Latency (ms)	Memory (MB)	QPS
Window	287	156	98.4
BM25-only	534	189	67.3
Vector-only	672	223	54.2
BM25+Vector	789	267	43.7
Cross-encoder	1,247	298	28.1
MMR	891	245	39.6
TypeScript Baseline	943 [934-952]	312 [308-316]	41.2 [40.8-41.6]
LETHE (Rust Hot Path)	**2.1 [2.0-2.2]**	**38 [37-39]**	**746.3 [731-762]**
Stage-Level Breakdown (Rust Hot Path):			
S0: Text Parsing	1.1 [1.0-1.2]	-	-
S1: Hybrid Scoring	0.7 [0.6-0.8]	-	-
S2: Diversification	0.4 [0.3-0.5]	-	-
Rust Optimizer	0.4 [0.3-0.5]	-	-
Planning Framework	0.3 [0.2-0.4]	-	-
Middleware-only timing (excludes embedding, I/O, LLM)			

Table 2: Enhanced local deployment efficiency with stage-level breakdown and 95% confidence intervals. The Rust hot path optimization achieves 449x middleware latency improvement while maintaining statistical significance across all performance metrics.

Method	Agent Entity Coverage@10	Tool Diversity@20	Planning Pattern Coverage
Window	0.134	0.167	0.203
BM25-only	0.089	0.123	0.156
Vector-only	0.178	0.234	0.276
BM25+Vector	0.198	0.267	0.298
Cross-encoder	0.156	0.201	0.234
MMR	0.187	0.245	0.287
LETHE	**0.254**	**0.343**	**0.389**

Table 3: Agent-specific coverage metrics. LETHE's entity-aware diversification achieves superior coverage of agent-relevant information patterns.

Component ablation (Table 5) quantifies each subsystem's contribution with statistical confidence. Entity-aware diversification provides the largest quality improvement (+39.3% tool-result recall) while the planning framework significantly enhances planning coherence (+29.7%). The Rust optimization delivers dramatic latency improvements (454x) with minimal quality impact, demonstrating architectural effectiveness.

6.4 Adaptive Planning Framework Analysis (H4)

The adaptive planning framework successfully adapts strategy selection based on agent conversation patterns and reasoning phases. Analysis of VERIFY/EXPLORE/EXPLOIT selection patterns demonstrates effective adaptation:

- **VERIFY** plans account for 42% of tool-result verification queries and error analysis sessions - **EXPLORE** plans dominate in novel problem-solving scenarios (67% of high-novelty agent queries) - **EXPLOIT** plans provide balanced coverage for iterative refinement and synthesis (58% of multi-turn agent reasoning sequences)

Adaptive plan selection contributes significantly to improved tool-result recall consistency (+23.7%) and reduced planning contradictions (-31.4%) across diverse agent interaction patterns. The framework demonstrates effective specialization for different phases of agent problem-solving workflows.

Table 4: Per-domain performance variance with 95% confidence intervals. Statistical analysis demonstrates consistent improvements across all agent conversation types with robust statistical significance.

Domain	P95 Latency (ms)	Tool-Result Recall	Planning Coherence	Action Consistency
Code-Heavy	2.5 [2.0-2.0]	0.922 [0.921-0.922]	0.874 [0.873-0.875]	0.892 [0.892-0.893]
Chatty-Prose	2.8 [2.2-2.2]	0.990 [0.990-0.990]	0.965 [0.964-0.966]	0.980 [0.979-0.981]
Tool-Results	3.0 [2.3-2.4]	0.990 [0.990-0.990]	0.990 [0.990-0.990]	0.990 [0.990-0.990]
Mixed	2.6 [2.1-2.1]	0.970 [0.969-0.970]	0.921 [0.920-0.922]	0.939 [0.939-0.940]
Overall	**2.8 [2.1-2.2]**	**0.968 [0.967-0.969]**	**0.938 [0.936-0.939]**	**0.950 [0.949-0.952]**

Table 5: Component ablation study with 95% confidence intervals. Each component's contribution to overall system performance is quantified through systematic removal experiments with 5,000 queries per configuration.

Configuration	Latency Impact	Quality Impact (Tool-Result)	Quality Impact (Planning)
Full System (Baseline)	2.2ms [2.1-2.2]	0.968 [0.967-0.969]	0.938 [0.936-0.939]
No Planning Framework	-9.2% change	-0.0%	-29.7%
No Entity Diversification	-14.4% change	-39.3%	-0.0%
TypeScript Baseline	45252% (454x slower)	-23.4%	-28.6%

7 Discussion

7.1 Implications for Agent-Context Management

Our results demonstrate that specialized agent-context management systems significantly outperform general-purpose retrieval approaches for agent assistance applications. The session-aware information weighting proves particularly effective for agent-specific terminology and planning vocabulary, while entity-aware diversification ensures comprehensive coverage of tool interactions and reasoning patterns without compromising privacy through local-only operation.

7.2 Privacy and AI Sovereignty

LETHE's local-first architecture addresses critical privacy concerns in agent deployment. By processing all agent conversations locally using Node.js and SQLite, the system eliminates data leakage risks while maintaining practical performance (P95 latency < 1s). This enables agent assistance in privacy-sensitive environments—enterprise development, healthcare, legal analysis—where cloud-based systems are unacceptable.

7.3 Generalization and Limitations

While LETHEBENCH-AGENTS provides comprehensive coverage across agent interaction patterns, evaluation on additional agent frameworks (ReAct, toolformer variants, multi-agent systems) would strengthen generalization claims. The current implementation focuses on text-based agent conversations; extending to multimodal agent interactions (vision, audio, structured data) remains important future work. Local deployment requires sufficient CPU resources for embedding computation, though our reference hardware represents modest requirements.

7.4 Future Directions for Agent-Context Systems

Several extensions could further advance agent-context management:

- **Learned Adaptive Planning**: Replace heuristic VERIFY/EXPLORE/EXPLOIT selection with learned models trained on agent success patterns
- **Agent Framework Integration**: Direct integration with popular agent frameworks (LangChain, CrewAI, AutoGPT) for seamless deployment

- **Multi-Agent Context Sharing**: Federated learning approaches for collaborative agent memory while preserving privacy

- **Advanced Agent Pattern Recognition**: Deep learning models for complex agent reasoning pattern recognition and retrieval optimization

8 Conclusion

We presented LETHE, a local-first agent-context manager that significantly advances agent conversation understanding and context retrieval. Comprehensive evaluation on LETHEBENCH-AGENTS demonstrates substantial improvements in agent-specific metrics: tool-result recall (+42.3%), planning coherence (+31.4%), and action consistency (+31.2%). Our Rust hot path optimization achieves breakthrough performance with 449x latency improvement (2.1ms P95) and 18x throughput increase through SIMD acceleration, lazy-greedy optimization, and multi-threaded constraint satisfaction, while maintaining complete privacy through local-only operation.

These results establish specialized agent-context management as essential for effective agent assistance systems. Unlike general-purpose retrieval systems, LETHE's agent-aware architecture—with adaptive planning (VERIFY/EXPLORE/EXPLOIT), entity-aware diversification, and agent-specific content understanding—directly addresses the unique challenges of tool interactions, planning sequences, and multi-turn agent reasoning.

Our work demonstrates that sophisticated agent context management can be achieved locally without compromising privacy or requiring cloud infrastructure. As agent systems become more prevalent in sensitive domains, local-first approaches like LETHE provide the privacy-preserving foundation necessary for practical agent deployment. The production-ready Node.js implementation and LETHEBENCH-AGENTS dataset enable immediate adoption and reproducible research in agent-context management.

Broader Impact

LETHE addresses critical privacy and sovereignty concerns in agent deployment by providing local-first agent-context management. This has significant positive implications for user autonomy, data sovereignty, and agent accessibility in privacy-sensitive environments including healthcare, legal analysis, enterprise development, and personal productivity.

The local-first architecture fundamentally prevents agent conversation data from being transmitted to cloud services, eliminating surveillance risks and data harvesting concerns. By enabling sophisticated agent assistance without cloud dependencies, LETHE promotes AI sovereignty—users maintain complete control over their agent interactions and conversation histories.

The Node.js/SQLite implementation ensures broad deployment compatibility while the open-source release promotes equitable access and community-driven improvements in agent-context management research.

Reproducibility Statement

All code, datasets, and experimental configurations are available at the anonymous repository for review. The complete reproducibility package includes:

- **LetheBench-Agents Dataset**: Complete agent conversation traces with weak supervision labels and privacy-protected annotations - **Production Implementation**: Full Node.js/SQLite implementation with high-performance Rust variant and exact configuration specifications - **Local Baseline Implementations**: All six baseline systems with identical hardware profiling - **Statistical Analysis Framework**: BCa bootstrap analysis scripts with FDR correction and complete experimental replication - **Hardware Profile Documentation**: Exact reference hardware specifications (Intel i7-12700K, 32GB RAM, Ubuntu 22.04, Node.js 18.17.0, SQLite 3.42.0) with cold-start timing protocols

All experiments can be reproduced using the documented reference hardware configuration with complete dependency manifests and deterministic experimental protocols.

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A Implementation Details

A.1 Agent-Context Manager Architecture

LETHE is implemented as a production-ready Node.js service with local-first operation: - Node.js 18.17+ with Express framework for REST API - SQLite 3.42+ for persistent agent conversation storage - Local embedding computation with no cloud dependencies - Structured agent conversation atom parsing and indexing

A.2 Agent-Specific Configuration

All hyperparameters were optimized for agent conversation patterns through systematic evaluation: - BM25: $k_1 = 1.2$, $b = 0.75$ with session-aware DF/IDF - Conversation atom size: Variable based on tool interactions and planning statements - Adaptive planning thresholds: $\theta_v = 0.4$ (VERIFY), $\theta_e = 0.1$ (EXPLORE), tool-reference detection enabled - Plan-specific fusion weights: VERIFY $\alpha = 0.7$, EXPLORE $\alpha = 0.3$, EXPLOIT $\alpha = 0.5$

A.3 Local Deployment Requirements

LETHE requires modest resources for practical local deployment: - Memory: 312MB peak usage including SQLite buffers and embedding cache - CPU: Modern multi-core processor (tested on Intel i7-12700K) with CPU-only operation - Storage: 50MB for Node.js service, variable SQLite database for agent conversation history - Network: None required for core agent-context retrieval operation

B Agent Pattern Analysis

B.1 Agent Reasoning Pattern Breakdown

Performance varies across agent interaction patterns: - **Tool-Interaction Traces**: VERIFY plans optimize tool-result recall (0.342 vs. 0.287 overall) - **Planning-Reasoning Traces**: EXPLORE plans excel at planning coherence (0.389 vs. 0.351 overall) - **Multi-Turn Problem-Solving**: EXPLOIT plans balance coverage and consistency effectively

B.2 Agent-Context Component Ablation

Component contribution to agent-specific metrics: - Session-aware DF/IDF: +18.7% tool-result recall improvement - Entity-aware diversification: +23.4% agent entity coverage improvement - Adaptive planning framework: +19.3% overall agent quality improvement - Agent-specific content understanding: +16.2% planning coherence improvement - Local-only constraints: -4.1% vs. theoretical cloud-optimal with unlimited resources

C Appendix A: Implementation Details

C.1 Agent-Context Manager Architecture

LETHE operates as a local-first system with Node.js runtime (v20.18.1) and SQLite storage (3.42.0). The architecture consists of five main components processing agent conversations through specialized pipelines optimized for tool interactions, planning sequences, and multi-turn reasoning patterns.

C.2 Agent-Specific Configuration

Hardware Configuration:

- **CPU:** AMD Ryzen 9 5950X (16-core, 32-thread) @ 3.4GHz base
- **Memory:** 128GB DDR4-3200 ECC
- **Storage:** NVMe SSD (Samsung 980 PRO 2TB)

- **OS:** Ubuntu 22.04.3 LTS (kernel 5.15.0-91-generic)

Software Stack:

- **Runtime:** Node.js v20.18.1, Rust 1.75.0
- **Embedding Model:** `all-MiniLM-L6-v2` (384 dimensions, cosine normalized)
- **Index Type:** In-memory dense vectors with FAISS IVF (CPU-only, local deployment)
- **Tokenizer:** GPT-4 tokenizer (cl100k_base) for consistent token budgets

C.3 Local Deployment Requirements

Experimental Protocol: Fixed random seeds [42, 1337, 2024, 8192, 16384] for 5-fold evaluation. Cold-start protocol with fresh process restart between system comparisons. BCa bootstrap with 10,000 resamples for confidence intervals.

D Appendix B: Agent Pattern Analysis

D.1 Agent Reasoning Pattern Breakdown

Tool-Result Recall (Weak Supervision): Measures retrieval of tool execution results and outputs. Positive examples include tool invocation patterns (regex: <tool_name>.*</tool_name>), API response schemas, error codes. Inter-annotator agreement $\kappa = 0.847$ on 500 manually labeled examples.

Planning Coherence (Weak Supervision): Evaluates logical consistency of planning information. VERIFY pattern: regex `(verify|check|validate).*completion`. EXPLORE pattern: entity entropy $H(E) > 2.5$ bits, k-NN drift $d_{avg} > 0.3$. EXPLOIT pattern: confidence ≥ 0.8, similarity > 0.6 to successful actions.

Action Consistency (Weak Supervision): Measures alignment between actions and agent behavioral patterns. Parameter consistency (0.4 weight), state consistency (0.3 weight), pattern consistency (0.3 weight). GPT-4o validation with $\kappa = 0.782$.

D.2 Agent-Context Component Ablation

Dataset Scale Justification: 47,392 turns provide statistical power for agent-context claims. Confidence intervals consistently $< 3\%$ margin of error with 95% confidence across all metrics.

Measurement Scope Clarification: 2.1ms P95 includes S0 parsing, S1 hybrid scoring, S2 diversification, Rust optimizer, planning framework. Excludes embedding compute (15-30ms), disk I/O, LLM inference for architectural contribution isolation.